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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CH614 – ORGANIC CHEMISTRY - IV

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The type of electronic transition possible in 2-methylpropane is
   a) σ-σ* b) π-π* c) n-π* d) n-σ*

2. Finger print region in IR spectroscopy corresponds to the
   a) 100-400 cm^{-1} b) 500-1500 cm^{-1} c) 1600-1800 cm^{-1} d) above 3000 cm^{-1}

3. The unit of coupling constant J in NMR spectroscopy is
   a) m^2 b) Nm^{-1} c) cal d) Hz

4. Total number of ^1H NMR signals for neo-pentane is
   a) 1 b) 2 c) 3 d) 5

5. Reduction of methyl isocyanides using $LiAlH_4$ will give
   a) Primary amine b) Secondary amine d) Tertiary amine d) None

6. The catalyst employed in clemmensen's reduction is
   a) $ZnCl_2$/HCl b) $NaBH_4$ c) Cl_2/KOH d) Zn(Hg)/HCl

7. Diels-Alder reaction is which type of addition
   a) 2+2 b) 4+4 c) 4+2 d) 1+4

8. Which is an alkaloid among the following
   a) Cellulose b) Pyridine c) Coniine d) Insulin

9. Nitration of furan occurs at position
   a) 2 b) 1 c) 3 d) 4

10. Skraup's synthesis is for the preparation of
    a) Aniline b) Naphthalene c) Quinoline d) Iso-quinoline

II. Fill up the blanks

11. λmax for 1,3-butadiene will be _____ than that of ethylene.

12. Characteristic peak for aldehydes in mass spectroscopy occurs at m/e = _____.

13. Hydroboration-oxidation of propene will give _____.

14. Treatment of an alkene with alkaline $KMnO_4$ gives _____.

15. Hydrogenation of pyridine using Raney Nickel will give _____.

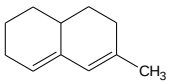
III. Match the following

16. Functional group	i) Reducing agent
17. Isotopic peak	ii) IR spectroscopy
18. Electronic transition	iii) Terpenes
19. Isoprene rulei	v) Mass spectroscopy
20. DIBAL	v) UV-Visible spectroscopy

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Calculate λmax using Woodward-Fieser rule for



22. Name any two examples for molecules exhibiting intra molecular hydrogen bonding.].

23. For an aromatic compound, if the λmax occurs at 255 nm, calculate its corresponding energy required for electronic transition.

24. Write the Larmor equation in ^1H NMR technique.

25. Define base peak. Give one example.

26. Write the fragmentation pattern for acetone in mass spectroscopy.

27. What happens when birch reduction is performed over benzoic acid? Give equation.

28. Give an example for clemmenson reduction.

29. What will happen if 1, 3-butadiene and acrolein undergo 4+2 cyclo addition?

30. Draw the structure of piperine.

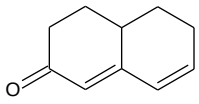
31. What is isoprene rule?

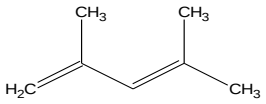
32. Give any one preparation for thiophene.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Calculate λmax using Woodward-Fieser rule for. **(2+2)**





b) How inter and intra molecular hydrogen bonding can be distinguished using IR spectroscopy. **(3)**

34. a) Write mass fragmentation pattern for toluene. **(4)**
b) Predict and draw the ^{1}H NMR spectrum for 2-bromo propane. **(3)**

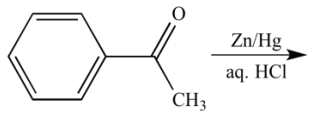
35. a) Predict the product formed when 2-butyne is hydrogenated using Linder's catalyst and birch reduction. **(2+2)**
b) Write Wolf-Kishner reduction for butanone and ethanol. **(1 ½ +1 ½)**

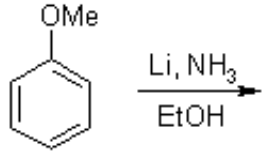
36. a) How are alkaloids extracted from plants? **(4)**
b) Draw the structure and IUPAC name for coniine. **(3)**

37. a) Write Fischer-indole synthesis. **(4)**
b) Write any three points about structural elucidation of menthol. **(3)**

38. a) Draw the structure of Pinene, Geraniol, Citral-A and Camphor. **(4)**
b) Write an example for Diels-Alder's reaction. **(3)**

39. a) Predict





(2+2)

b) Give the preparation and two uses of pyridine. **(1+2)**

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